

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester II Examination, 2013-14 (Ancillary)

Paper No. MTB-AM-203: Ancillary Mathematics I

Time: 3 Hours

Full Marks: 70

Code: BS/Sem II/4

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Which of the sets are equal: (i) $\{x : x \text{ is a letter in 'follow'}\}$; (ii) $\{x : x \text{ is a letter in 'wolf'}\}$; (iii) $\{x : x \text{ is a letter in 'flow'}\}$.
- (b) If $A = \{3, 4\}$ and $B = \{1, 2, 5\}$, find $A \times B$.
- (c) If $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$, $C = \{3, 4, 5, 6\}$, find $(A \cap B) \cap C$.
- (d) Which of the following is a function $f : X \rightarrow X$ for $X = \{1, 2, 3, 4\}$: $f_1 = \{(2, 3), (1, 4), (2, 1), (3, 2), (4, 4)\}$? (State reason.)
- (e) $R = \{(x, y) : x + 2y = 5\}$ on $\mathbb{N}$. Is R transitive?
- (f) Find k if $\begin{vmatrix} 5 & k+2 \\ k-2 & 3 \end{vmatrix} = \begin{vmatrix} k+3 & 4 \\ -k & 2 \end{vmatrix}$.
- (g) Find AB where $A = \begin{bmatrix} 3 & 2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.
- (h) If $A = \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$, find $A - 2B$.
- (i) Without expanding the determinant, prove $\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} x & y & z \\ p & q & r \\ a & b & c \end{vmatrix}$.
- (j) Determine the rank of $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$.
- (k) If $n(A \cup B) = 20$, $n(A \cap B) = 8$, $n(B) = 12$, find $n(A)$.

Question 2.

- (a) Show $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$.
[6]
- (b) In a survey: 1051 used bus, 709 car, 1252 train, 406 bus and car, 316 bus and train, 301 train and car, 51 all three. How many were surveyed?
[6]

Question 3.

(a) Evaluate $\begin{vmatrix} 167 & 19 & 21 \\ 39 & 13 & 14 \\ 81 & 24 & 26 \end{vmatrix}$. [6]

(b) Show $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$. [6]

Question 4.

(a) Solve by determinants: $3x + 5y - 7z = 13$, $4x + y - 12z = 6$, $2x + 9y - 3z = 20$. [6]

(b) (i) Find x if $\begin{vmatrix} 2x & 0 \\ x & x \end{vmatrix} = \begin{vmatrix} 10 & 0 \\ -1 & 2 \end{vmatrix}$. (ii) Solve for X : $X \begin{pmatrix} 2 & 5 \\ 1 & 4 \end{pmatrix} = \begin{pmatrix} -6 \\ 0 \end{pmatrix}$. [3+3]

Question 5.

(a) If $A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 1 & 2 \\ 7 & 2 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 3 & 0 \\ -9 & 1 & 2 \end{pmatrix}$, compute $A^2 - 4B^2$. [6]

(b) If $A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 3 & 2 \\ 1 & -1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 0 & 0 \\ 5 & 0 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 4 \\ 4 \\ 1 \end{pmatrix}$, compute $(AB)C$ and $A(BC)$ and verify $(AB)C = A(BC)$. [6]

Question 6.

(a) Find the inverse of $A = \begin{pmatrix} 0 & 1 & -2 \\ -1 & 0 & 5 \\ 2 & -5 & 0 \end{pmatrix}$. [6]

(b) Find the rank of $A = \begin{pmatrix} 0 & 2 & 2 & -1 \\ 1 & 0 & 4 & 1 \\ 3 & 1 & 0 & 9 \\ 1 & 1 & -2 & 9 \end{pmatrix}$. [6]

Question 7.

(a) Solve: $x + 3y - 2z = 0$, $2x - y + 4z = 0$, $x - 11y + 14z = 9$. [6]

(b) Test consistency and solve: $x + 2y + 3z = 14$, $3x + y + 2z = 11$, $2x + 3y + z = 11$. [6]

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